

New York University Tandon School of Engineering
Computer Science and Engineering
CS-GY 9223I: Final Exam Review.

Logistics

- **Time/place:** 3:20pm - 5:30pm on Wednesday 12/18 in our usual classroom. It will be a 1.5 exam, but you can take up to 2 hours.
- **Material allowed:** One double sided sheet of paper with whatever you want on it is allowed.

Concepts to Know

Random variables and concentration.

- This was the first section of the course, but the tools we learned never stopped being important. You will not be tested on them in isolation, but don't be surprised if problems require applying techniques like computing expectations/variances or random variables, applying the basic concentration bounds, applying union bounds, etc.

Convex + non-convex optimization.

- Meaning of condition number for convex optimization problem. Meaning in the special case of $f(x) = \|Ax - b\|_2$.
- Idea and goal of preconditioning.
- Coordinate descent. How does runtime compare to gradient descent for $f(x) = \|Ax - b\|_2$.
- Stationary point definition.
- Convergence of gradient descent to stationary point for non-convex function – what assumptions are needed.

Singular Value Decomposition

- Singular value decomposition.
- Relation to norms: $\|A\|_F^2 = \sum_{i=1}^d \sigma_i^2$. $\|A\|_2^2 = \max_{x: \|x\|_2=1} \|Ax\|_2^2 = \sigma_1^2$.
- Optimal low-rank approximation.
- When are matrices low-rank?
- Power method and its analysis.
- Semantic embeddings from similarity measure.
- The Courant-Fischer min-max principal. We stated this for eigenvalues, but recall that the singular values of a matrix $\mathbf{A}$ are eigenvalues of the matrix

Spectral Graph Theory

- Matrix representations of graphs (adjacency matrix $\mathbf{A}$, Laplacian matrix $\mathbf{L}$, edge-vertex incidence matrix $\mathbf{B}$) $\mathbf{A}^T \mathbf{A}$. so the statement adapts to singular values.
- Intuitively what is the meaning of the smallest Laplacian eigenvector.
- Stochastic block model.
- Basic strategy for recovering the partition in a stochastic block model.
- Matrix concentration inequalities (you don't need to know the one stated in class, but know what it means to write down the expectation of a random matrix and bound its distance from that expectation).
- Davis-Kahan eigenvector perturbation theorem.

Randomized Numerical Linear Algebra

- What is a matrix sketch.
- Subspace embedding theorem.
- How subspace embeddings can be used to approximately solve least squares regression.
- ϵ -net arguments. Why was one needed for subspace embedding proof? Even if you can't reproduce the proof yourself, know the main components.
- Size of ϵ -net for the unit ball in $\mathbb{R}^d$.
- You do not need to know Randomized Hadamard matrices or the Fast JL algorithm.

Compressed Sensing + High Dimensional Geometry

- What is the compressed sensing problem and why is it important.
- Restricted Isometry Property, why it allows for sparse recovery.
- What matrices satisfy RIP? With what parameters? How many rows?
- Basis pursuit optimization problem for efficient recovery (you don't need to be able to reproduce the analysis).
- Concentration of volume around the surface and equators of a high dimensional ball.
- There are exponentially many nearly orthogonal vectors in high dimensional space.

Practice Problems

From *Foundations of Data Science* book (<https://www.cs.cornell.edu/jeh/book.pdf>). Note that this book uses $\|x\|$ to denote $\|x\|_2$ for a vector x .

- **Exercises:** 3.6, 3.7, 3.8, 3.10, 3.11, 3.12, 3.13, 3.18 (I stated in class, make sure you can prove), 3.20, 3.21, 3.22, 3.26, 7.27, 12.31, 12.36
- **Exercises:** 12.33, 12.38 (try to figure out what the eigenvectors and eigenvalues of A are by hand)
- **Exercises:** 6.16. Understand the second equation on page 194: $\mathbb{E}[X] = AB$.
- **Exercises:** 2.2
 1. For $V \in \mathbb{R}^{n \times d}$ with orthonormal columns and vector $x \in \mathbb{R}^n$, when is $\|V^T x\|_2 = \|x\|_2$? Always, sometimes, or never?

2. Let A_k be the optimal k -rank approximation for A obtained via an SVD. prove that $\|A - A_k\|_2 = \sigma_{k+1}(A)$.
3. For any $V \in \mathbb{R}^{n \times d}$ with orthonormal columns, VV^T is the projection matrix onto the subspace spanned by the columns of V (V 's column span). We used this fact many times when discussing low-rank approximation. Show that $VV^T = (VV^T)(VV^T)$. Why does this property make intuitive sense if VV^T is a projection?
4. Prove that $\|A - A_k\|_F^2 = \|A\|_F^2 - \|A_k\|_F^2$. Is the same statement true if we switch these to spectral norms $\|\cdot\|_2$?
5. Let $X \in \mathbb{R}^{n \times 900}$ have random entries drawn independently as $\{0, 1\}$, each with probability $1/2$. Let $Y \in \mathbb{R}^{n \times 900}$ have rows corresponding to 30×30 pixel black and white images of handwritten digits. All entries of Y are in $\{0, 1\}$. How do you expect $\sum_{i=1}^{30} \sigma_i(X)^2$ and $\sum_{i=1}^{30} \sigma_i(Y)^2$ to compare?
6. You have a data matrix $X \in \mathbb{R}^{n \times d}$ where each row corresponds to a student and each column corresponds to their grade on an assignment. The final column is their cumulative grade. What do you expect the rank of this matrix to be? Do you think it is well approximated by an even lower rank matrix?
7. Let $X \in \mathbb{R}^{n \times d}$ have SVD $U\Sigma V^T$ with singular values $\sigma_1(X), \dots, \sigma_d(X)$. What are the eigenvalues of $(X^T X)^q$? What are its eigenvectors. How about $(X X^T)^q$. What is the runtime required to apply either of these two matrices to a vector?
8. In the stochastic block model, why is clustering with the second largest eigenvector of the expected adjacency matrix equivalent to clustering with the second smallest eigenvector of the expected Laplacian? Are these two approaches identical when clustering using the actual rather than the expected matrices? Describe a natural variant of the stochastic block model where these two algorithms would not be equivalent even on the expected matrices.